

TrES-1b

R-0262

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_170615 · created 2026-07-17T09:04:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457919.7734 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.039
Transit depth (Rp/Rs)^2	1.32 +/- 0.89 [%]
Orbital Inclination (inc)	85.17 +/- 1.47
Airmass coefficient 1 (a1)	0.979 +/- 0.042
Airmass coefficient 2 (a2)	0.028 +/- 0.041
Scatter in the residuals of the lightcurve fit is	4.31 %
Best Comparison Star	#2 - [390, 228]
Optimal Method	PSF photometry
Transit Duration (day)	0.053 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.115 ± 0.039 vs 0.1358 (deviation 0.37σ)
Duration fit vs published	0.053 d vs 0.1042 d (deviation 1.12σ)
Mid-time O–C	-4.3 min
Depth SNR	2.1
Residual scatter	4.31 %

Light curve

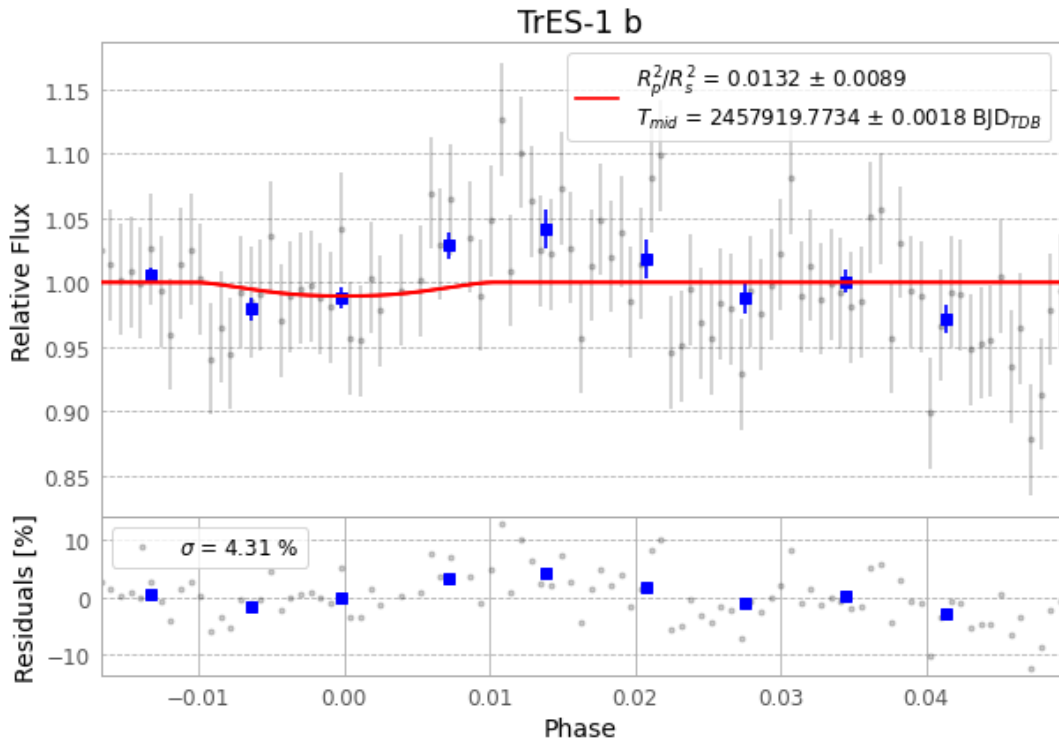


Figure 1 — FinalLightCurve_TrES-1 b_2017-06-14.png (SHA-256 62ac9f39e341a579...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 113], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 087a38eca7200b69...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 7dbc7ef

Data provenance

97 FITS frames (64 MB), TRES-1170615051512.FITS → TRES-1170615100312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-170615.log (1249f409fa6f...), FinalLightCurve_TrES-1 b_2017-06-14.png (62ac9f39e341...), FinalLightCurve_TrES-1 b_2017-06-14.csv (f31660ac25ec...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 09:04Z.